ch23 Electromagnetism

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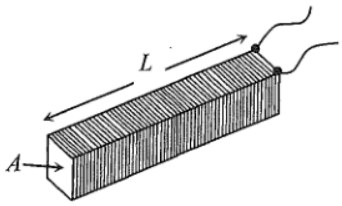
28.

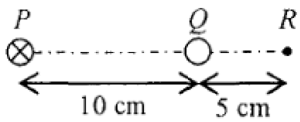


The figure shows a closely packed long solenoid of cross-sectional area A and length L having a total of N turns. If the solenoid carries a constant direct current throughout, which of the following changes can increase the magnetic flux density B at its central cross-section ?

	length	**cross-sectional area**	**total number of turns**
A.	$2L$	$2A$	$2N$
B.	L	$2A$	N
C.	$2L$	A	N
D.	L	A	$2N$

2021 Q25

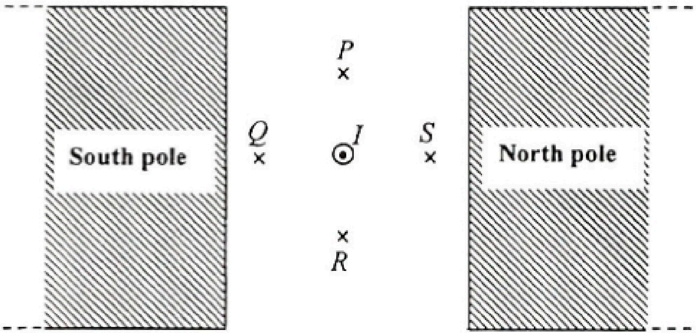
25. In the figure below, PQR is a straight line with $PQ = 10$ cm and $QR = 5$ cm. A long straight wire carrying a current of 0.3 A (directed into the paper) is placed at P. When another long straight wire carrying a current I is placed at Q, the resultant magnetic field at R becomes zero. Determine the direction and magnitude of I.



	direction of I	magnitude of I
A.	into the paper	0.1 A
B.	into the paper	0.9 A
C.	out of the paper	0.1 A
D.	out of the paper	0.9 A

2018 Q27

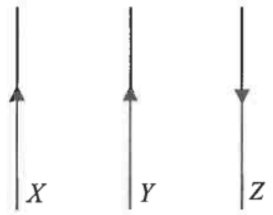
27. A straight wire carrying a current I pointing out of the paper is placed in a uniform magnetic field between two pole pieces as shown. At which point, P, Q, R or S, can the resultant magnetic field be zero ? Neglect the effects of the Earth's magnetic field.



A. P
B. Q
C. R
D. S

2022 Q27

27. The figure below shows three parallel wires carrying currents in the directions shown.

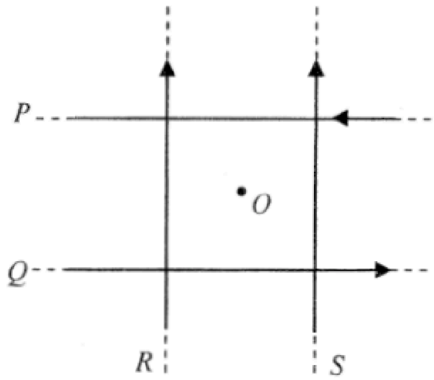


If one of the wires experiences zero resultant magnetic force, that wire

A. must be *X*.
B. must be *Y*.
C. must be *Z*.
D. may be *Y* or *Z*.

2013 Q27

27.

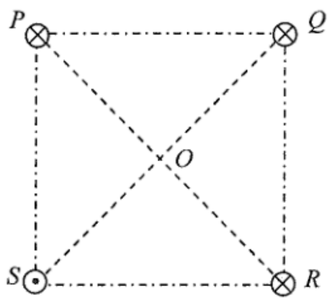


In the figure, four long straight wires P, Q, R and S in the same plane carry equal currents in the directions shown. The wires are insulated from each other. O is a point on the same plane and is equidistant from each wire. Removing which wire would increase the magnetic field strength at O ?

A. wire P
B. wire Q
C. wire R
D. wire S

2014 Q26

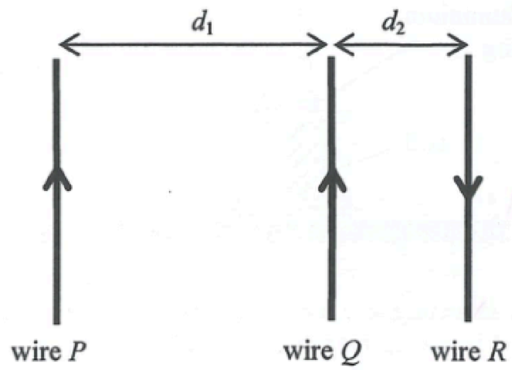
26.



Four long straight parallel wires P, Q, R and S carrying currents of equal magnitude are situated at the vertices of a square as shown. P, Q and R each carries a current directed into the paper while S carries a current directed out of the paper. The direction of the resultant magnetic field at the centre O of the square is along

A. OP.
B. OQ.
C. OR.
D. OS.

26. Three parallel wires P, Q and R are arranged with separations d_1 and d_2 (with $d_1 > d_2$) as shown. Each wire carries the same magnitude of current flowing in the directions indicated. If the magnitude of the magnetic force per unit length exerted on Q by P is F, what are the direction and magnitude of the **resultant magnetic force** per unit length exerted on Q ?

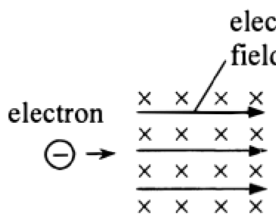


	direction of resultant magnetic force exerted on Q	**magnitude of resultant magnetic force per unit length exerted on Q**
A.	to the right	greater than $2F$
B.	to the right	smaller than F
C.	to the left	greater than $2F$
D.	to the left	smaller than F

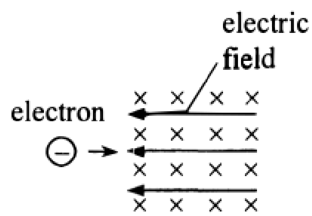
2012 Q30

30. An electron enters a region in which both a uniform electric field E and a uniform magnetic field B exist. The magnetic field B is pointing into the paper. In which direction should the electric field be applied so that the electron could be undeflected ?

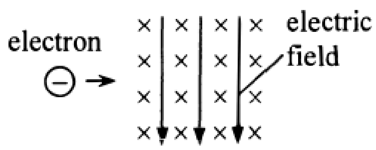
A.



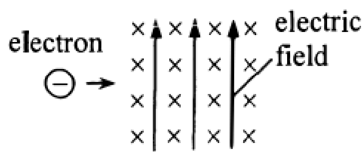
B.



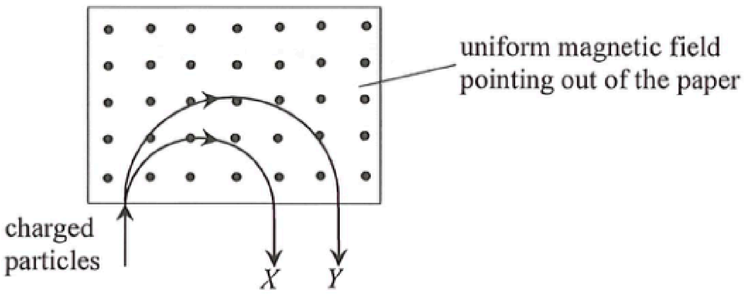
C.



D.



*30. The figure shows two charged particles X (mass m_X) and Y (mass m_Y) which enter a region of uniform magnetic field pointing out of the paper. X and Y, having the **same speed and charge**, move along the paths in the plane of the paper as shown below.

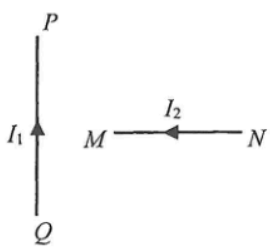


X and Y are

A. positively charged and $m_X > m_Y$.
B. positively charged and $m_X < m_Y$.
C. negatively charged and $m_X > m_Y$.
D. negatively charged and $m_X < m_Y$.

2024 Q26

26.

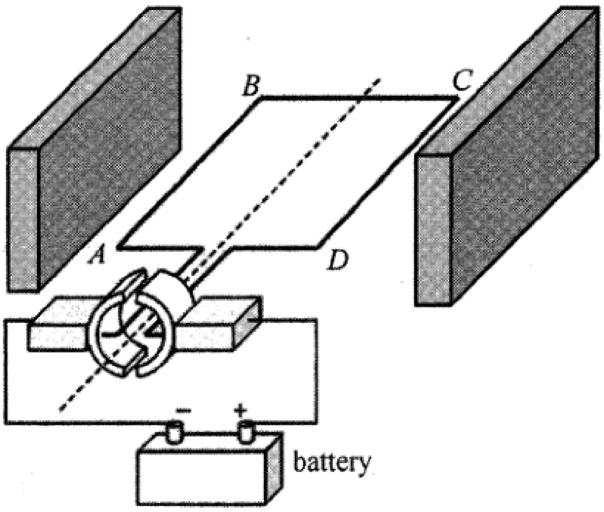


PQ in the figure is a section of a fixed long straight wire carrying a current I_1. MN is a section of another long straight wire carrying a current I_2. MN is perpendicular to PQ and both wires are in the same plane. What is the direction of the magnetic force acting on MN due to PQ ? Neglect the effects of the Earth's magnetic field.

A. pointing out of the paper

2017 Q26

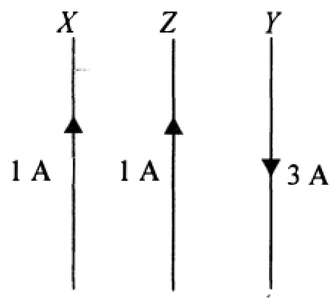
26. The figure shows a simple d.c. motor, the coil $ABCD$ is mounted between the poles of two slab-sha magnets.



Which of the following statements is correct ?

A. The turning effect is zero when the coil is vertical.
B. The magnetic force acting on BC is the greatest when the coil is horizontal.
C. The direction of the magnetic force acting on AB remains constant.
D. The direction of the current in the coil remains unchanged.

29. In the figure below, X, Y and Z are three long straight parallel wires with Z placed midway between X and Y. X and Z carry currents of 1 A in the same direction while Y carries a current of 3 A in the opposite direction. The magnetic force per unit length experienced by wire X due to wire Z is of magnitude F.

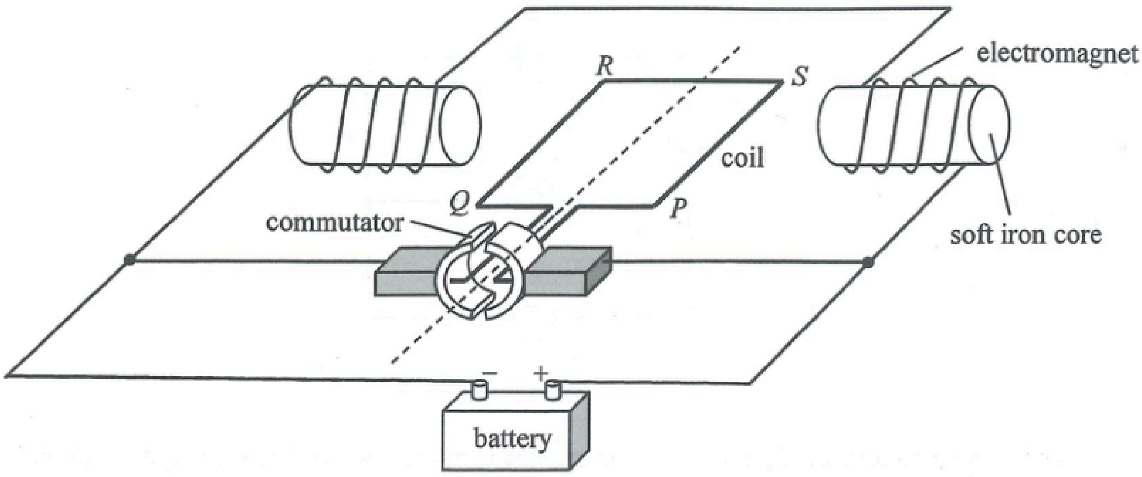


The magnetic force per unit length acting on wire Z due to both X and Y is

A. $2F$ to the right.
B. $2F$ to the left.
C. $4F$ to the right.
D. $4F$ to the left.

2019 Q27

27.

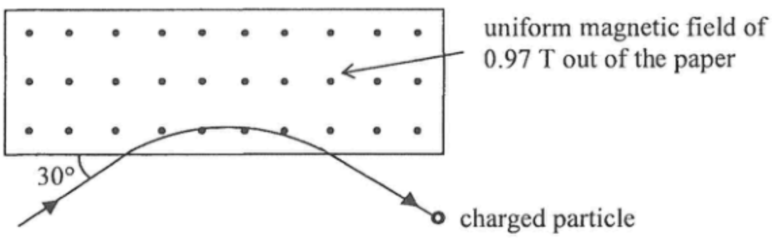


The figure shows the structure of a motor. The coil *PQRS* and the two electromagnets are connected to a battery so that the coil rotates continuously. If a sinusoidal a.c. source of frequency 50 Hz is used instead of a battery, the coil will

A. remain at rest.
B. oscillate at a frequency 50 Hz.
C. rotate to a vertical position and then stop.
D. rotate continuously.

2024 Q29

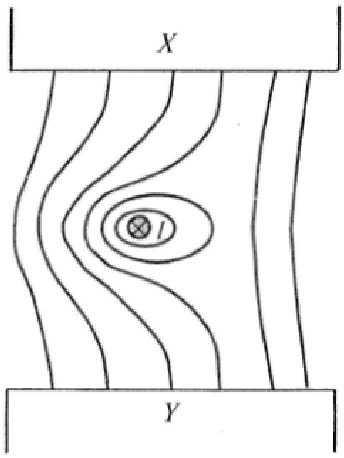
*29.



A charged particle enters and leaves a magnetic field of uniform flux density 0.97 T along the path as shown. It experiences a magnetic force of 5.9×10^{-12} N within the magnetic field. Given that the speed of the particle is 1.9×10^{7} m s^{-1}, find its charge.

A. -3.2×10^{-19} C
B. $+3.2 \times 10^{-19}$ C
C. -6.4×10^{-19} C
D. $+6.4 \times 10^{-19}$ C

26.

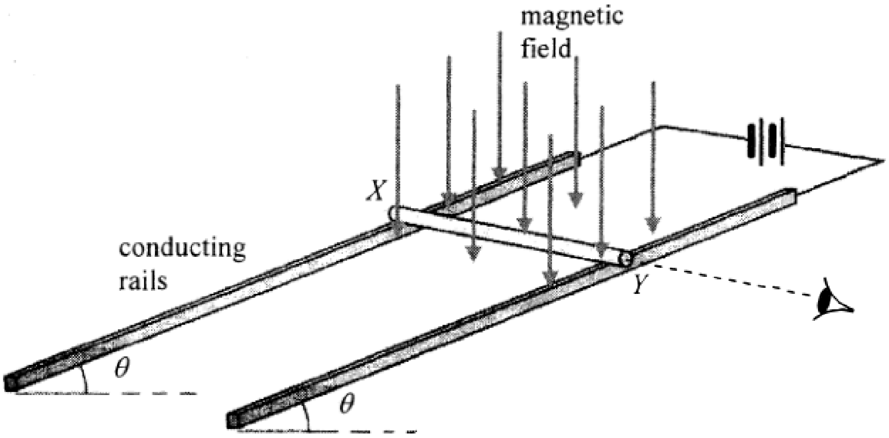


A straight wire carrying current I pointing into the paper is placed in a magnetic field between pole pieces X and Y. The figure shows the resultant field line pattern. What is the polarity of pole piece X and in what direction is the magnetic force acting on the wire ? Ignore the effect of the Earth's magnetic field.

	polarity of X	direction of magnetic force
A.	**N**	to right
B.	**N**	to left
C.	**S**	to right
D.	**S**	to left

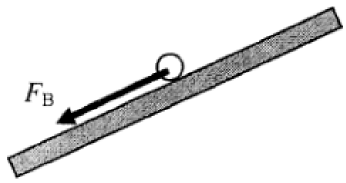
2013 Q28

28.

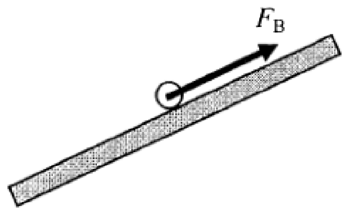


A copper rod XY is placed on a pair of smooth inclined conducting rails which are located in a magnetic field applied vertically downward. The rails make an angle θ to the horizontal and a battery is connected to the rails as shown above. Which diagram shown below represents the magnetic force F_B acting on the rod when viewed from end Y?

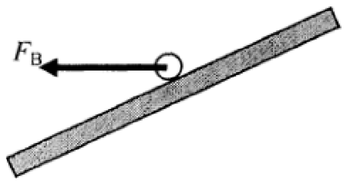
A.



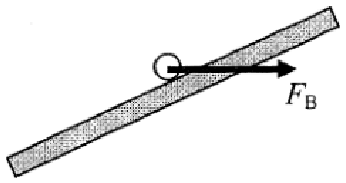
B.



C.

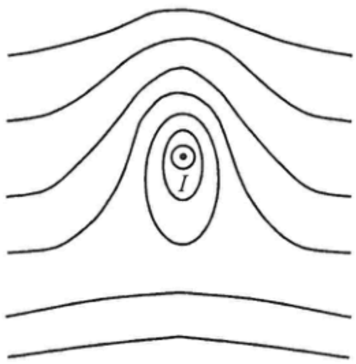


D.



2020 Q26

26. The figure below shows the magnetic field pattern on a horizontal surface around a long vertical straight wire carrying a steady current I pointing out of the paper. The Earth's magnetic field is **NOT** neglected.



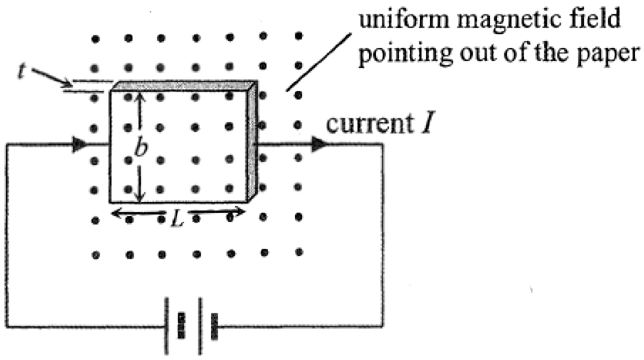
Top view

What are the directions of the following ?

	the horizontal component of the Earth's magnetic field	the magnetic force experienced by the current-carrying wire
A.	$\leftarrow$	$\downarrow$
B.	$\leftarrow$	$\uparrow$
C.	$\rightarrow$	$\downarrow$
D.	$\rightarrow$	$\uparrow$

2015 Q30

*30.

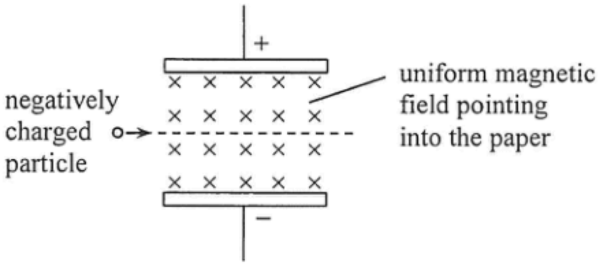


A slice of conductor (with a certain resistivity) of thickness t, breadth b and length L is connected to a battery of constant e.m.f. and negligible internal resistance. A steady current I passes through it as shown. When a uniform magnetic field is applied normally to the slice, a Hall voltage V is set up between two of its opposite faces. If the thickness and the breadth of the slice are reduced to ½ t and ½ b respectively, what would the following quantities be ?

	current passing through the slice	**Hall voltage developed**
A.	$\frac{I}{4}$	$\frac{V}{4}$
B.	I	$\frac{V}{4}$
C.	$\frac{I}{4}$	$\frac{V}{2}$
D.	I	$\frac{V}{2}$

2020 Q27

*27.



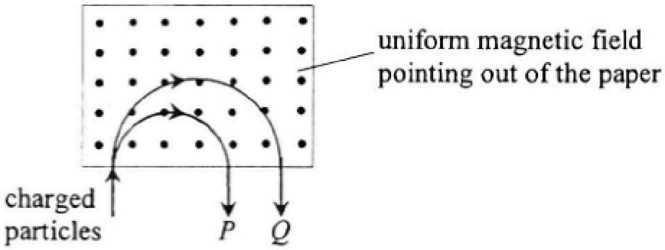
A negatively charged particle goes undeflected through a region in which a uniform electric field and a uniform magnetic field are set up as shown. The electric field is set up by the potential difference across the two parallel metal plates. Which of the following changes may cause the charged particle to deflect downward ? Neglect the effects of gravity.

(1) increasing the potential difference across the plates
(2) increasing the magnitude of the charge on the particle
(3) increasing the particle's speed entering the region

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2018 Q28

*28.



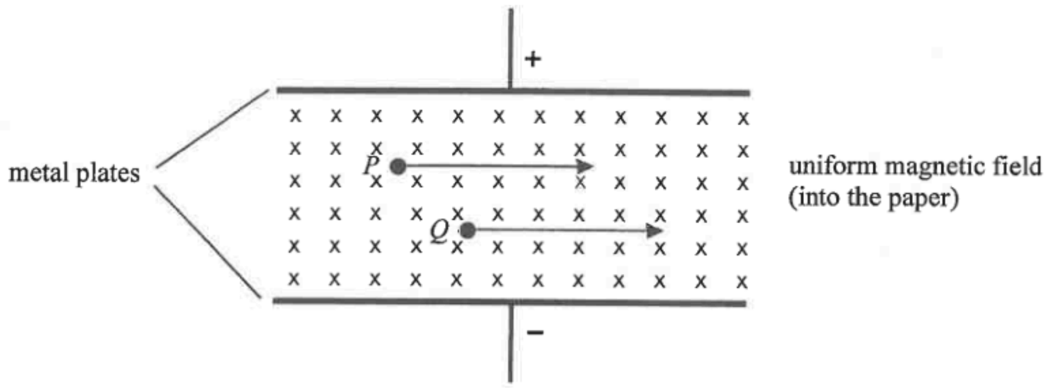
P and Q are two particles carrying the same amount of charge but of different masses. They travel with the same speed and enter a uniform magnetic field pointing out of the paper as shown. Semi-circular paths with different radii are described before they emerge from the field. Which descriptions below are correct ?

(1) Both P and Q are positively charged.
(2) P and Q emerge from the field with the same speed.
(3) The mass of Q is greater than that of P.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2022 Q29

*29. Charged particles P and Q are moving in a region with mutually perpendicular uniform electric and magnetic fields as shown.



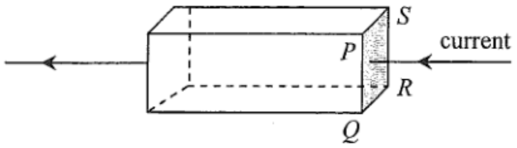
If both particles are not deflected by the fields, which of the following statements **must be** correct ? Neglect the effects of gravity.

(1) They are positively charged.
(2) They are moving with the same velocity.
(3) They have the same charge to mass ratio.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

2014 Q29

*29.

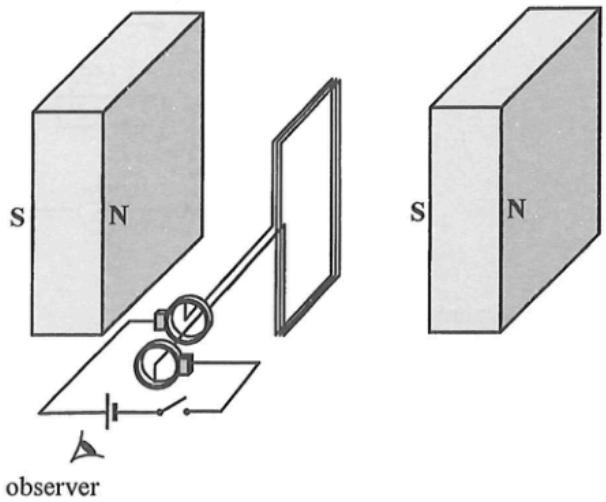


The figure shows a current flowing from right to left in a metal block with cross-section *PQRS*. When a uniform magnetic field is applied to the block, the electric potential of the side *PQ* is found to be higher than that of the side *SR*. In which direction can the magnetic field be applied ?

A. from *P* to *Q*
B. from *Q* to *P*
C. from *P* to *S*
D. from *S* to *P*

2025 Q28

28. The figure shows a rectangular coil connected in series with a dry cell and a switch through a pair of slip rings. The magnetic field between the two magnets is uniform and is perpendicular to the plane of the coil at the moment shown.

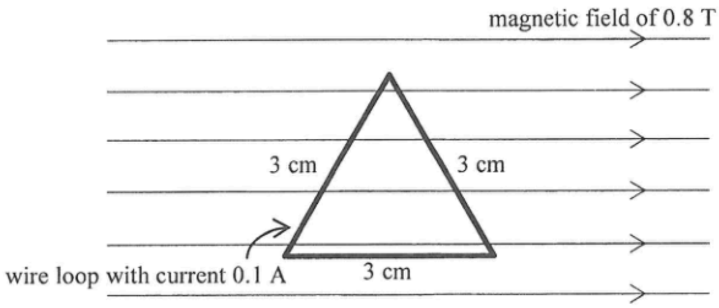


When the switch is closed, the coil will

A. remain at rest.
B. rotate in a clockwise direction continuously.
C. rotate in an anticlockwise direction continuously.
D. change the direction of rotation after turning through 180°.

2026 Q27

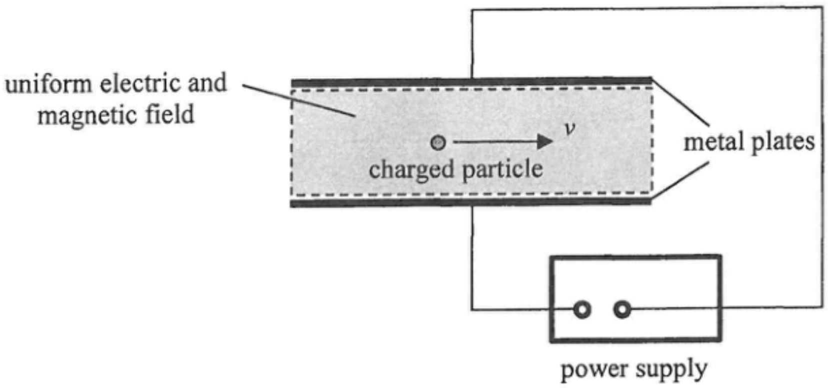
27.



An equilateral triangular wire loop with sides of 3 cm is placed in a uniform magnetic field of 0.8 T. One of the sides of the loop is parallel to the field as shown. A current of 0.1 A flows through the loop. Find the net magnetic force acting on the loop.

A. 0 N
B. 2.8×10^{-3} N
C. 5.6×10^{-3} N
D. 7.6×10^{-3} N

*29.



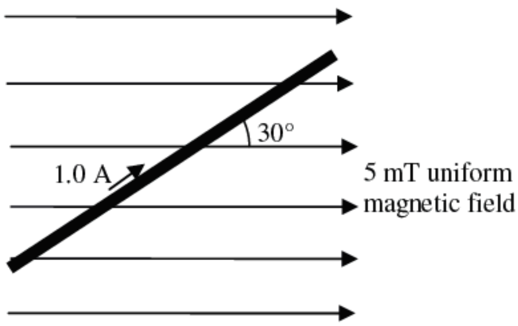
A particle with charge q travels horizontally at speed v without deflection through a region of uniform electric and magnetic field as shown. The electric field is set up by the potential difference across the two parallel metal plates. The magnetic field B is applied perpendicular to the paper. The whole set-up is in vacuum. Which of the following combinations allow the particle to travel horizontally without deflection through the region ? Neglect the effects of gravity.

	charge of particle	**speed of particle**	**magnetic field**
(1)	q	$2v$	$\frac{B}{2}$
(2)	$-q$	v	B
(3)	$2q$	v	B

A. (1) and (2) only
B. (2) and (3) only
C. (1) and (3) only
D. (1), (2) and (3)

pp Q31

31. The figure below shows a current of 1.0 A flowing in a metal rod of length 0.5 m. The rod is placed inside a region with a uniform magnetic field of strength 5 mT. What is the direction and the magnitude of the magnetic force acting on the rod ?



Direction **Magnitude**

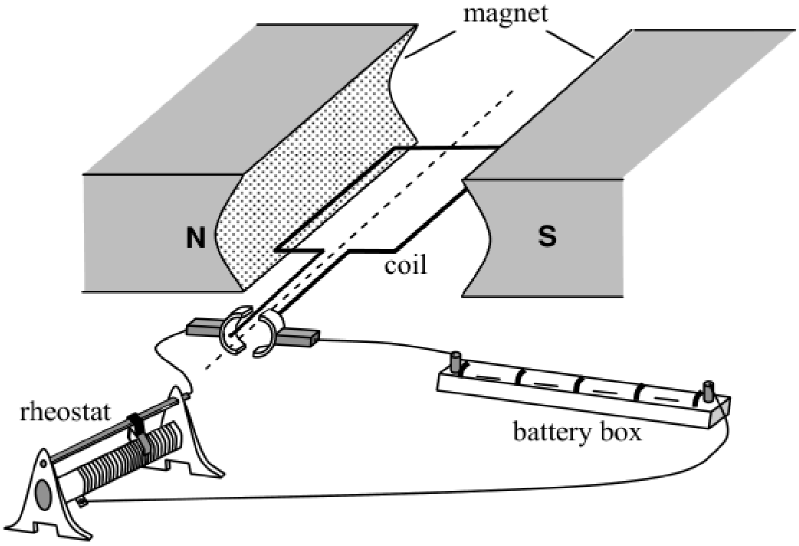
pp Q32

*32. A Hall probe is placed in a uniform magnetic field. The slice of semiconductor inside the Hall probe is 1.3×10^{-3} m thick and has 10^{25} charge carriers per cubic metre. When a steady current of 0.4 A passes through the slice, a Hall voltage of 2×10^{-5} V is set up. What is the magnetic field strength detected by the probe ? Assume that the magnitude of the charge of each charge carrier is 1.6×10^{-19} C.

A. 0.104 T
B. 0.962 T
C. 1.04 T
D. 9.62 T

sap Q28

28.



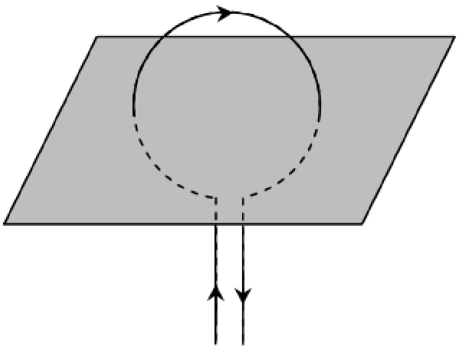
The figure shows a simple motor. Which of these changes would increase the turning effect of the coil ?

(1) using a stronger magnet
(2) reducing the resistance of the rheostat
(3) using a coil with a smaller number of turns

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

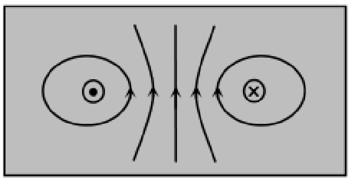
sap Q29

29.

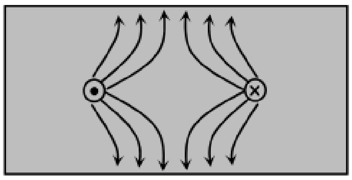


Which diagram shows the magnetic field pattern around a flat circular current-carrying coil, in the plane shown ?

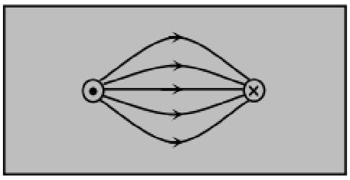
A.



B.



C.



D.